From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA TEST SERIES

## ANSWERS, HINTS & SOLUTIONS

## FULL TEST – II
## PAPER-2

| *Q.NO* | PHYSICS | CHEMISTRY | MATHEMATICS |
|---|---|---|---|
| 1. | A | C | A |
| 2. | B | C | A |
| 3. | C | B | B |
| 4. | B | A | A |
| 5. | C | B | A |
| 6. | A | A | A |
| 7. | D | A | B |
| 8. | B | B | A |
| 9. | D | B | A |
| 10. | A | B | A |
| 11. | A | A | B |
| 12. | B | A | C |
| 13. | C | C | C |
| 14. | C | D | B |
| 15. | B | C | C |
| 16. | A | D | C |
| 17. | B | A | D |
| 18. | A | D | B |
| 19. | A | B | A |
| 1. | (A) → (q, r), (B) → (q, r, s)<br>(C) → (p), (D) → (p) | (A) → (r), (B) → (p)<br>(C) → (s), (D) → (q) | (A) → (p), (B) → (q)<br>(C) → (r), (D) → (s) |
| 2. | (A) → (r), (B) → (p, q, r, s)<br>(C) → (q), (D) → (s) | (A) → (p), (B) → (q)<br>(C) → (r), (D) → (s) | (A) → (r), (B) → (s)<br>(C) → (q), (D) → (p) |
| 3. | (A) → (s)<br>(B) → (r)<br>(C) → (p)<br>(D) → (q) | (A) → (p, q, r, s)<br>(B) → (p, q, s)<br>(C) → (r)<br>(D) → (p, r) | (A) → (s)<br>(B) → (p)<br>(C) → (r)<br>(D) → (q) |

# Physics

PART – I

3. According to the question

$$\frac{g}{\left(1+\frac{h}{R_e}\right)^2} = g\left(1-\frac{h}{R_e}\right)$$

$$-\frac{h}{R_e}+1-\frac{h^2}{R_e^2}=0$$

$$h = \frac{\left(\sqrt{5}-1\right)R_e}{2}$$

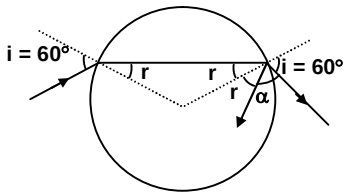
4. From figure it is clear

$\alpha = 180-(i+r)$

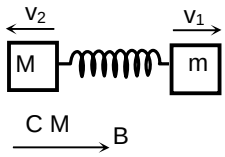
and $\mu = \frac{\sin i}{\sin r} \Rightarrow \sqrt{3} = \frac{\sin 60°}{\sin r}$

$r = 30°$

$\alpha = 90°$




6.




16. $F = qvB \sin\theta = qvB\ (\theta = 90°)$

So, $B = \frac{F}{qv} = \frac{3.2\times10^{-20}}{1.6\times10^{-19}\times4\times10^{3}} = 5\times10^{-7}\ T$

Now, $\frac{\mu_0}{2\pi}\left[\frac{I_p}{5}+\frac{I_q}{2}\right] = 5\times10^{-7}$

$\Rightarrow I = 4$ amp.

If the distance of point R from third current carrying current is X, then

$B_R = 0$

$\frac{\mu_0}{4\pi}\frac{2\times2.5}{|x|}+5\times10^{-7}=0$

so $x = \pm 1$ m

19. $P(2\pi r)dr = 2\pi r\ell\eta\ \frac{dv}{dr} - 2\pi(r+dr)\ \ell\eta\ \frac{dv}{dr}$

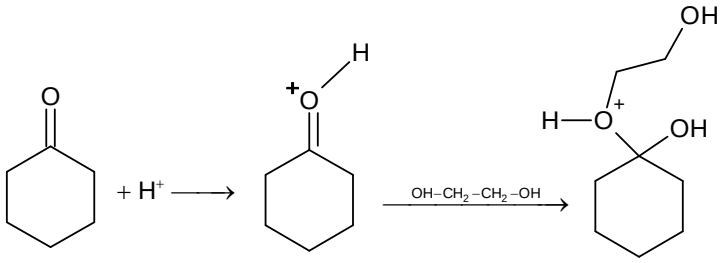
$P(2\pi r)dr = -2\pi\ \ell\eta\left(\frac{dv}{dr}\right)dr$

$P(2\pi r) = -2\pi\ \ell\eta\left(\frac{-2v_0 r}{R^2}\right)$

$P = \frac{2\eta\ell v_0}{R^2}$

# Chemistry

PART – II

1. 


2. (III) is most reactive (resonance activation) followed by N (inductive activation). (II) is more deactivated (resonance deactivation) followed by (I) (inductive deactivation).

3. $\lambda \propto \frac{1}{\sqrt{KE}}$

$= \sqrt{\frac{KE_1}{KE_2}} = \frac{\lambda_2}{\lambda_1}$

$= \frac{0.99\lambda_1}{\lambda_1}$

$KE_1 = (0.99)^2 KE_2$

$KE_2 \approx 1.02\ KE_1$

% change is KE = $\frac{KE_2 - KE_1}{KE_1} \times 100$

$\approx 2\%$

4. $\Delta H > 0, \Delta S > 0 \Rightarrow$ Reaction may be non-spontaneous at $25^oC$

$\Delta G = \Delta H - T\Delta S$

$= 180 - 298 \times 150 \times 10^{-3}$

$= 135.3 > 0$

= Non-spontaneous

To make it spontaneous $(\Delta G < 0)$. We have to increase the temperature.

$T = \frac{\Delta H}{\Delta S} = \frac{180 \times 10^3}{150}$

$= 1200\ K = 927^oC.$

11. Above critical temperature ($T_C$), gas can not be liquified on cooling, the average energy of molecule decreases.

12. Positive charge on nitrogen of diazo group is stabilized by electron releasing group.

13. Within amino acid, proton is accepted from $-COOH$ group by $-NH_2$ group to from "$^- COO - R - \overset{+}{N}H_3$".

14. More the number of alkyl substitute at double bond, greater its thermodynamic stability.

15. C – H bond is broken in non rate determining step, therefore, substitution of $\alpha$-H by deuterium doesn't affect the rate of reaction.

16. There are total four types of $\alpha$-H and two types of carbonyl, hence a total of eight aldol would be formed.

17. $E^{o}_{cell} = 2.37 + 0.8$
= 3.17 V
$E^{o} = \frac{0.059}{2} \log K_c$
$\log K_c = 107.45$

18. Maximum work = nFE
= $6 \times 10^2$ kJ

# Mathematics PART – III

1. Perfect square = $\left[\sqrt{100}\right] - 1 = 9$(excluding one)

Perfect cubes = $\left[100^{1/3}\right] - 1 = 3$

Perfect $4^{th}$ powers = $\left[100^{1/4}\right] - 1 = 2$

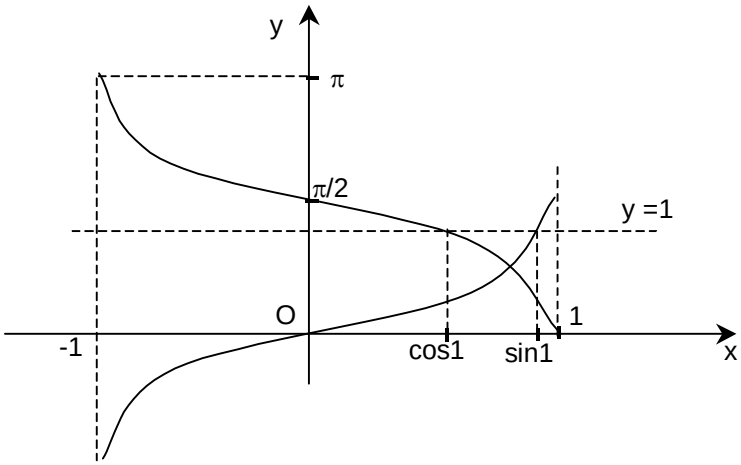
Perfect $5^{th}$ powers = $\left[100^{1/5}\right] - 1 = 1$

Perfect $6^{th}$ powers = $\left[100^{1/6}\right] - 1 = 1$

Now, perfect $4^{th}$ powers have already been counted in perfect squares and perfect $6^{th}$ powers have been counted with perfect squares as well as with perfect cubes

Hence the total ways = 9 + 3 + 1 – 1 = 12

2. $[\sin^{-1}x] > [\cos^{-1}x] \Rightarrow x > 0$




Clearly $[\cos^{-1}x] = \begin{cases} 1, & x \in [0, \cos 1] \\ 0, & x \in (\cos 1, 1] \end{cases}$

$[\sin^{-1}x] = \begin{cases} 0, & x \in [0, \sin 1) \\ 1, & x \in [\sin 1, 1] \end{cases}$

Hence $[\sin^{-1}x] > [\cos^{-1}x] \Rightarrow x \in [\sin 1, 1]$.

3. As roots are of opposite sign, product of roots < 0

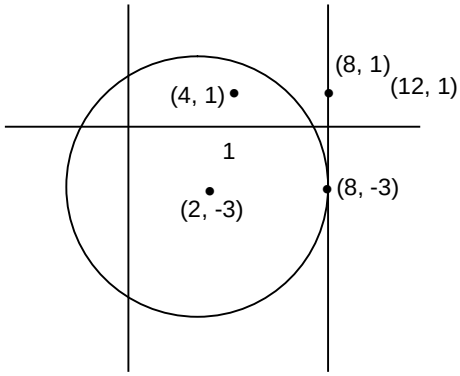
$\Rightarrow a^2 + b^2 - 1 < 0$

$\Rightarrow a^2 + b^2 < 1$

$\Rightarrow |a + ib| < 1$

So, the point a + ib lies inside a circle of centre (0, 0) and radius 1.

4. $|z - 2 + 3i| = 6$ is a circle with centre (2, –3) and radius 6 and $|z - 4 - i| = |z - 12 - i|$ is the perpendicular bisector of (4, 1) and (12, 1) is a line parallel to axis imaginary. Now this line
The line is tangent to circle at complex number (8 – 3i). Hence only one complex number satisfies the above equation.




6. Co–ordinates of point T (a cos $\alpha$, 0) so distance from focus of the point T is a ($e \pm \cos\alpha$)

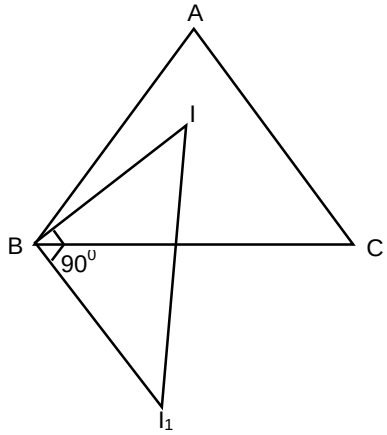
7. There are 8 even and 9 odd numbers. So probabilities of getting first even number is $\frac{8}{17}$ and probabilities of getting second odd number = $\frac{9}{17}$, so required probabilities = $\frac{8}{17}\times\frac{9}{17} = \frac{72}{729}$

8. $BI = r \operatorname{cosec}\frac{B}{2} = 4R \sin\frac{A}{2}\sin\frac{C}{2}$

$BI_1 = r_1 \sec\frac{B}{2} = 4R \sin\frac{A}{2}\cos\frac{C}{2}$

$\Rightarrow II_1 = \sqrt{(BI)^2 + (BI')^2} = 4R \sin\frac{A}{2}$

$\Rightarrow \sum II_1 = 4R \sum \sin\frac{A}{2} \Rightarrow R = \frac{15}{8}$




9. $y^r = \left(1+\frac{1}{r}\right)\left(1+\frac{2}{r}\right)\left(1+\frac{3}{r}\right).....\left(1+\frac{n-1}{r}\right)$

$\Rightarrow \log y = \frac{1}{r}\sum_{p=1}^{n-1}\log\left(1+\frac{p}{r}\right)$

$\lim_{n\to\infty} \log y = \int_0^k \log(1+x)\ dx = (k+1)\log(1+k) - k$

11. $px^2 + 4\mu xy + qy^2 + 4a(x + y + 1) = 0$ represents pair of straight lines iff
$4apq + 16a^2\mu - 4a^2p - 4a^2q - 16a\mu^2 = 0$
$\Rightarrow 4\mu^2 - 4a\mu + ap + aq - pq = 0.$
For real $\mu$, $16a^2 - 4.4(ap + aq - pq) \geq 0$
$\Rightarrow (a - p)(a - q) \geq 0 \Rightarrow a \leq p$ or $a \geq q$

12. $f(x) = ax^2 - bx + 2$
$f(0) = 2$
$f(-1) = a + b + 2 < 0$ ($\because$ $a + b + 4 < 0$)

$\therefore$ f(0) f(– 1) < 0
$\therefore$ One root lie between (– 1, 0)
Nothing can be said about ab

13. Here only one condition is given. So degree is 2.

14. Any point on the parabola is $(t^2, t)$. Shift the origin to (–4, 0) so that the line becomes X + y = 0 and the point $(t^2, t)$ becomes $(4 + t^2, t)$ where X = x + 4. If $(X_1, y_1)$ is the image of $(4 + t^2, t)$ in X + y = 0, then $\begin{bmatrix} X_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}\begin{bmatrix} 4+t^2 \\ t \end{bmatrix}$

$\Rightarrow X_1 = -t$, $y_1 = -(4 + t^2)$ and in the original coordinates $x_1 = -4 - t$, $y_1 = -4 - t^2$
$\Rightarrow$ the equation of image is $(x + 4)^2 = -(y + 4)$.

15. Shift the origin to the point (0, 3) so that any point $(x_1, y_1)$ on the reflected line is given by $\begin{bmatrix} x_1 \\ y_1 - 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} \frac{1-q(h+3)}{p} \\ h \end{bmatrix}$ (since m = 0)

$\Rightarrow x_1 = \dfrac{1-q(h+3)}{p}$, $y_1 - 3 = -h \Rightarrow px_1 = qy_1 + 1 - 6q$

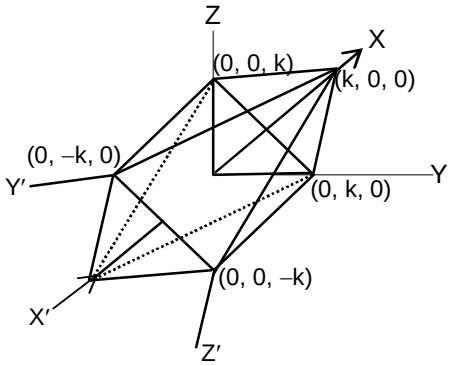
and hence the reflected line is px – qy = 1 – 6q.

16. Equation of the tangent to the given circle at (2, – 1) is 2x + y – 3 = 0.
Shift the origin to the point (–2, 0) so that the two lines becomes y = X and 2X + y – 7 = 0. Any point on the line is (h, 7 – 2h) and its reflection in y = X is given by $\begin{bmatrix} X_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} h \\ y-2h \end{bmatrix}$

$\Rightarrow X_1 = 7 - 2h$, $y_1 = h \Rightarrow X_1 = 7 - 2y_1$
$\Rightarrow x_1 + 2 = 7 - 2y_1$ and hence the reflection of the tangent is
2y + x – 5 = 0 or $y = \dfrac{x}{-2} - \dfrac{5}{4}(-2)$ which touch the parabola $y^2 = -5x$.

17. In the new definition $l = \dfrac{x}{|x| + |y| + |z|}$, etc.

18. d(O, P) = k
$\Rightarrow |x| + |y| + |z| = k$
which represents a set of 9 planes making intercepts of lengths k on positive as well negative sides of all three axes. See the adjacent figure.




19. The maximum value of d(O, P) = circumradius of sphere = a.

## SECTION – B

1. **(A)** $I = \int \frac{dx}{a^2 - x^2} = \frac{x}{a^2 - x^2} - \int \frac{2xdx}{(a^2 - x^2)^2} x = \frac{x}{a^2 - x^2} + 2\int \frac{a^2 - x^2 - a^2}{(a^2 - x^2)^2} dx$

$= \frac{x}{a^2 - x^2} + 2I - 2a^2 \int \frac{dx}{(a^2 - x^2)^2}$

$\Rightarrow \int \frac{dx}{(a^2 - x^2)^2} - \frac{1}{2a^2} \int \frac{dx}{a^2 - x^2}$

$= \frac{x}{2a^2(a^2 - x^2)} + c$ .

**(B)** Put $x = a \sec \theta$

$\Rightarrow dx = a \sec \theta \tan \theta \, d\theta$

$\therefore I = \int \frac{a \sec \theta \tan \theta d\theta}{a^2 \sec^2 \theta \cdot a \tan \theta} = \frac{1}{a^2} \sin \theta + c = \frac{\sqrt{x^2 - a^2}}{a^2 x} + c$ .

**(C)** Put $x = a \sin\theta$

$\Rightarrow dx = a \cos\theta \, d\theta$

$\therefore I = \int \frac{a \cos\theta a \cos\theta d\theta}{a^2 \sin^2 \theta} = \int \cot^2 \theta d\theta = -\cot\theta - \theta + c$

$= -\frac{\sqrt{a^2 - x^2}}{x} - \sin^{-1} \frac{x}{a} + c = -\frac{\sqrt{a^2 - x^2}}{x} + \cos^{-1} \frac{x}{a} + c$ .

**(D)** Put $x = a \sec \theta$

$\Rightarrow dx = a \sec \theta \tan \theta \, d\theta$

$\therefore I = \int \frac{a \tan\theta a \sec\theta \tan\theta d\theta}{a \sec\theta} = a \int \tan^2 \theta d\theta$

$= a \tan\theta - a\theta + c$

$= \sqrt{x^2 - a^2} - a \sec^{-1} \frac{x}{a} + c$ .

2. **(A)** Required area $= 2\int_0^4 \sqrt{x} dx - \frac{1}{2} \times 4 \times 4$

$= \frac{4}{3}[x^{3/2}]_0^4 - 8 = \frac{8}{3}$

**(B)** $\int_0^{\pi/2} (\sin^5 x + \cos^5 x) dx$

$= 2\int_0^{\pi/2} \sin^5 x dx = 2\left[\frac{4}{5} \times \frac{2}{3}\right] = \frac{16}{15}$

**(C)** For equation $S + k = 0$ to represent pair of lines

$$\begin{vmatrix} 1 & 2 & -2 \\ 2 & 3 & 1 \\ -2 & 1 & 1+k \end{vmatrix} = 0$$

$\Rightarrow 3(1 + k) - 1 - 2(2 + 2k + 2) - 2(2 + 6) = 0$

$\Rightarrow k = -22$

**(D)** Let p.v. of given points be $A(\hat{i}+\hat{j}+\hat{k})$, $B(2\hat{i}+2\hat{j}+\hat{k})$ and $C(3\hat{i}-\hat{k})$, so that two vectors in the plane may be $\overrightarrow{AB}=\hat{i}+\hat{j}$ and $\overrightarrow{AC}=2\hat{i}-\hat{j}-2\hat{k}$

Thus, $\begin{vmatrix}\alpha & 2 & 1\\ 1 & 1 & 0\\ 2 & -1 & -2\end{vmatrix}=0$

$\Rightarrow -2\alpha - 2(-2) + (-1-2) = 0$

$\Rightarrow \alpha = \frac{1}{2}$

3. **(A)** The equation will have roots of opposite sign if it has real roots and product of roots is negative $4(b^2+1)^2 - 12(b^2-3b+2) \geq 0$ and $\frac{b^2-3b+2}{3} < 0$

$\Rightarrow 1 < b < 2$

**(B)** The probability of problem being solved is $1 - P(\bar{A})\cdot P(\bar{B})\cdot P(\bar{C})$

$= 1 - \left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)(1-\alpha)$

$= \frac{2}{3}+\frac{\alpha}{3} \in \left(\frac{2}{3},\ 1\right)$

**(C)** $x = 5 - (y+z)$

$yz + x(y+z) = 8$

$yz + (y+z)(5-(y+z)) - 8 = 0$

$y^2 + y(z-5) + (z^2-5z+8) = 0$

For real solution, $(z-5)^2 - 4(z^2-5z+8) \geq 0$

$(z-1)\left(z-\frac{7}{3}\right) \leq 0$

$1 \leq z \leq \frac{7}{3}$